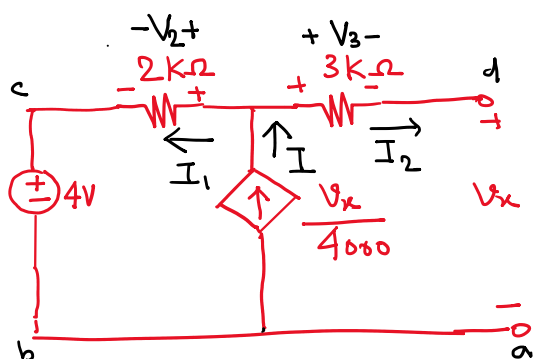
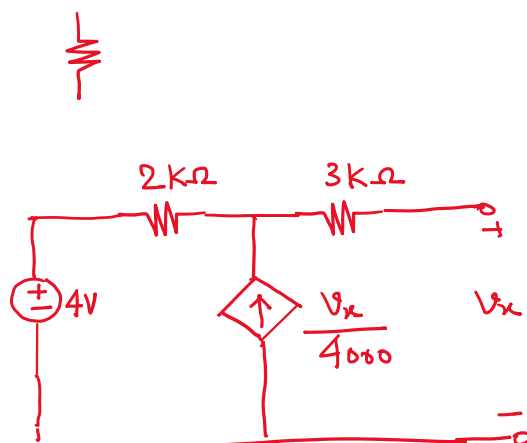
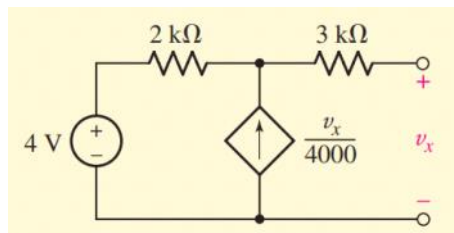


Thevenins_Dependent_source

Saturday, December 7, 2024 7:17 PM



$$I = I_1 + I_2, \quad I_2 = 0 \text{ [open]}$$

$$I = I_1 = \frac{v_x}{4000}$$

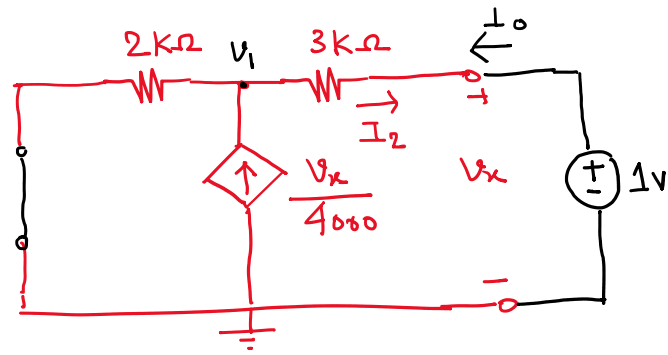
Applying KVL in the loop $a \rightarrow b \rightarrow c \rightarrow d$

$$+4 + V_2 - V_3 - v_x \Rightarrow 4 + 2I_1 - 3I_2 + v_x = 0$$

$$\Rightarrow 4 + 2 \times 10^3 \times \frac{v_x}{4000} - 3 \times 0 + v_x = 0$$

$$\Rightarrow v_x = 8V = V_{oc}$$

For R_{TH} . Remove the independent source and Add a 1V source as seen ~~in~~ in the CKT.



Apply node Analysis: at node 1

$$\frac{v_1}{2 \times 10^3} - \frac{v_x}{4000} + \frac{v_1 - 1}{3 \times 10^3} = 0$$

$$\Rightarrow \frac{v_1}{2000} - \frac{1}{4000} + \frac{v_1 - 1}{3000} = 0$$

$$\Rightarrow \frac{6v_1 - 3 + 4v_1 - 4}{12000} = 0$$

$$\Rightarrow 10v_1 = 7 \Rightarrow v_1 = \frac{7}{10} \text{ V}$$

$$I_2 = \frac{v_1 - 1}{3 \times 10^3} = \frac{7/10 - 1}{3 \times 10^3} = \frac{-3/10}{3 \times 10^3} = -\frac{1}{10} \text{ mA}$$

$$R_{TH} = \frac{V_0}{I_0} = \frac{1}{-I_2} = \frac{1}{1/10} = 10 \text{ K}$$